$$\hat{q}_i(\tau) = \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}\tau}\hat{q}_i\mathrm{e}^{-\mathrm{i}\hat{H}_{\mathcal{E}}\tau} = \sqrt{\frac{1}{2m_i\omega_i}}\left(\hat{a}_i\mathrm{e}^{-\mathrm{i}\omega_i\tau} + \hat{a}_i^{\dagger}\mathrm{e}^{\mathrm{i}\omega_i\tau}\right). \tag{5.9}$$